ASSIGNMENT 1

Experiment 1 Name: Interfacing LED with Arduino UNO using Digital Pin.

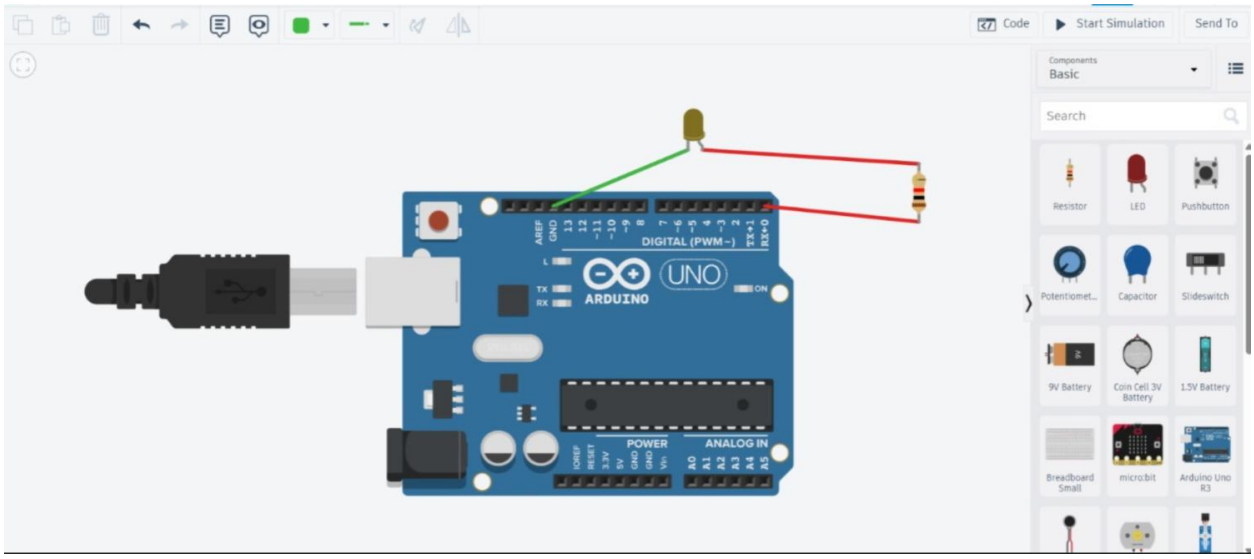
Apparatus / Requirements: Arduino UNO**,** LED, 220Ω Resistor, Connecting Wires**,** USB Cable, Computer/Laptop.

Hardware: Arduino UNO
Software: Tinkercad

Theory: Digital pins of Arduino UNO (0–13) are used for input and output operations. When configured as OUTPUT, they provide HIGH (5V) or LOW (0V) signals to control external devices such as LEDs.

Working Principle: When the digital pin is set to HIGH, current flows through the resistor and LED, causing it to glow. When set to LOW, the LED turns OFF. The resistor limits excess current.

RESULTS:



1 (Arduino Uno R3)

```
// C++ code
//
void setup()
{
  pinMode(0, OUTPUT);
}

void loop()
{
  digitalWrite(0 , HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(0 , LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Conclusion: The experiment demonstrates successful LED interfacing using a digital pin of the Arduino UNO.

Experiment 2 Name: Interfacing LED with Arduino UNO using Analog Pin.

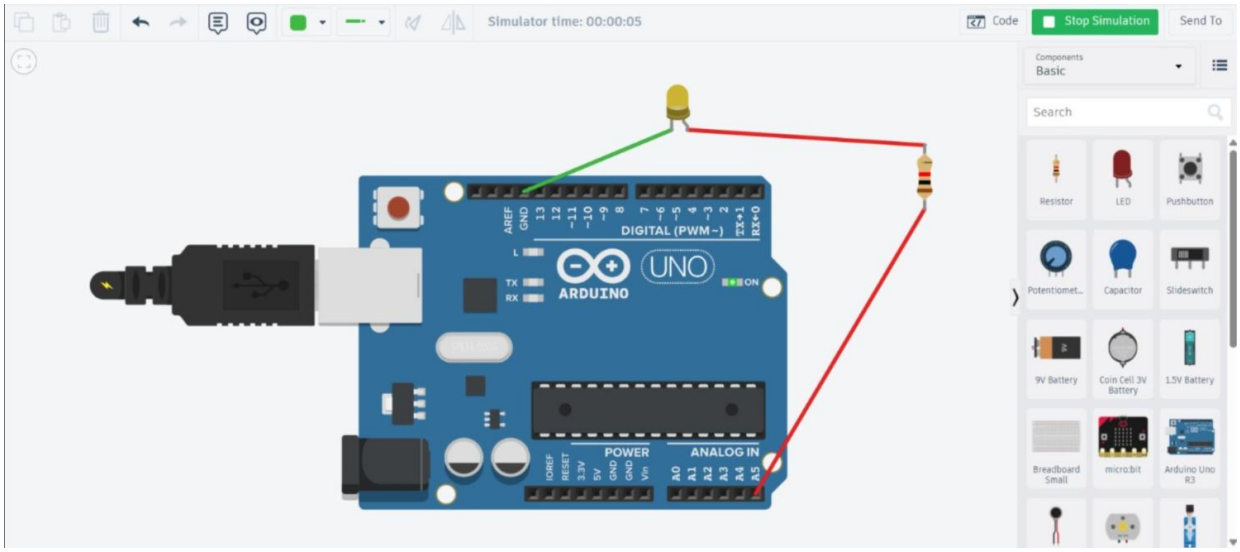
Apparatus / Requirements: Arduino UNO, LED, 220Ω Resistor, Connecting Wires, USB Cable, Computer/Laptop.

Hardware: Arduino UNO
Software: Tinkercad

Theory: Analog pins of Arduino UNO (A0–A5) are primarily used for analog input. However, they can also function as digital input/output pins. When configured as OUTPUT, they provide HIGH (5V) or LOW (0V) signals to control external devices such as LEDs.

Working Principle: When the analog pin (A5) is set as OUTPUT and written HIGH, current flows through the resistor and LED, causing it to glow. When set to LOW, the LED turns OFF. The resistor limits excess current.

RESULTS:



1 (Arduino Uno R3)

```
// C++ code
//
void setup()
{
  pinMode(A5 , OUTPUT);
}

void loop()
{
  for (int i = 0; i<255; i++)
  {
    analogWrite(A5,i);}
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Conclusion: The experiment verifies that Arduino analog pins can also function as digital output pins for controlling an LED.

ASSIGNMENT 2

Experiment Name: Interfacing an LDR Sensor with Arduino UNO through Analog Pin.

Components Required: Arduino UNO, LDR Sensor, 10kΩ/220Ω Resistor, Connecting Wires, USB Cable, Computer/Laptop.

Hardware Used: Arduino UNO
Simulation Platform: Tinkercad

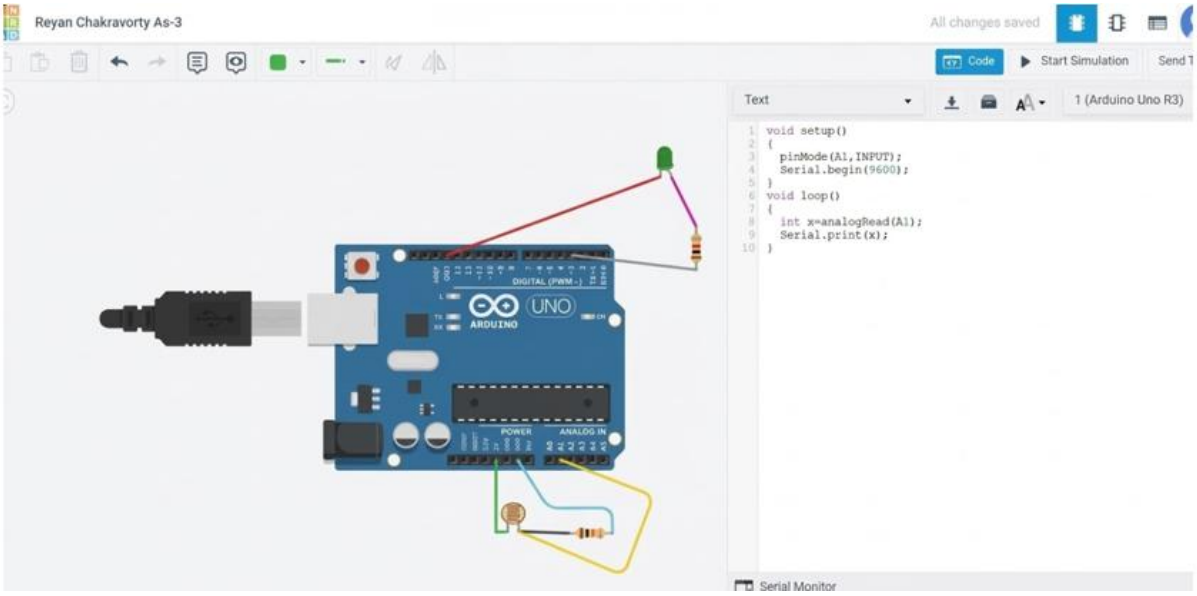
Theory

The Arduino UNO contains analog input pins (A0–A5) that allow it to read varying voltage signals from sensors. A Light Dependent Resistor (LDR) is a type of resistor whose resistance changes depending on the intensity of light falling on it. When the amount of light increases, the resistance of the LDR decreases, and when light decreases the resistance increases.

Working Principle

In this circuit, the LDR and a resistor are connected to form a voltage divider. The output voltage from this divider is given to analog pin **A1** of the Arduino. The Arduino reads the voltage using the **analogRead()** function and converts it into a value between **0 and 1023**. These values represent the light intensity detected by the LDR.

Result



Conclusion

This experiment confirms that an LDR can be interfaced with Arduino UNO to measure changes in light intensity using analog input pins.

ASSIGNMENT 3

Experiment Name: Controlling a Servo Motor using Arduino UNO PWM Signal.

Components Required: Arduino UNO, Servo Motor, Jumper Wires, USB Cable, Computer/Laptop.

Hardware Used: Arduino UNO
Simulation Platform: Tinkercad

Theory: A servo motor is a device that allows accurate control of rotational position. It works based on **Pulse Width Modulation (PWM)** signals generated by a microcontroller. The Arduino UNO provides multiple PWM pins such as **3, 5, 6, 9, 10, and 11**, which can be used to control servo motors.

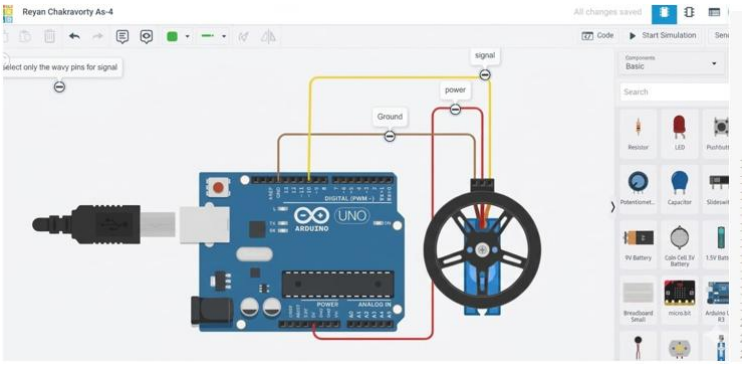
A typical servo motor contains three wires:

- **VCC (Power)** connected to 5V
- **Ground (GND)** connected to ground
- **Signal wire** connected to a PWM pin of Arduino

Working Principle

The Arduino generates PWM pulses through a digital PWM pin which are interpreted by the servo motor. The duration of these pulses determines the rotation angle of the servo shaft. By varying the pulse width, the servo motor can move between different angles such as **0°, 90°, and 180°**.

Result




```
#include<Servo.h>
Servo s;
void setup()
{
  s.attach(10);
}
void loop()
{
  for(int i=0;i<180;i++)
  {
    s.write(i);
    delay(100);
  }
  for(int i=179;i>=0;i--)
  {
    s.write(i);
    delay(100);
  }
}
```

Conclusion: The experiment successfully demonstrates the control of a servo motor using PWM signals generated by the Arduino UNO.

ASSIGNMENT 4

Experiment Name: Controlling an LED using a Push Button with Arduino UNO.

Components Required: Arduino UNO, LED, Push Button Switch, 220Ω Resistor, 10kΩ Resistor, Connecting Wires, USB Cable, Computer/Laptop.

Hardware Used: Arduino UNO
Simulation Platform: Tinkercad

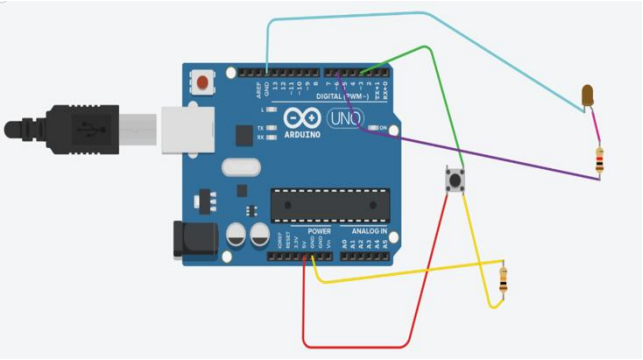
Theory

Arduino digital pins can be used both as **input and output** pins. A push button acts as a simple input device which sends a signal to the Arduino when pressed. The Arduino reads this signal and performs a specific action such as turning an LED ON or OFF.

Working Principle

The push button is connected to a digital input pin along with a resistor to maintain a stable voltage level. When the button is pressed, the Arduino receives a **HIGH signal**. Based on this input, the Arduino activates the LED connected to the output pin. When the button is released, the signal becomes **LOW** and the LED turns OFF.

Result




```
void setup()
{
  pinMode(3, INPUT);
  Serial.begin(9600);
  pinMode(6, OUTPUT);

}

void loop()
{
  int x =digitalRead(3);
  Serial.print(x);
  if(x==1)
  {
    digitalWrite(6,HIGH);
  }
  else
  {
  digitalWrite(6,LOW);
  }

}
```

Conclusion

The experiment verifies that a push button can be used as an input device to control an LED through Arduino UNO.